

BHUVANESHWAR 7K

727624 BADO49

AI&DS - A (II)

NATURAL LANGUAGE PROCESSING SYSTEMS.

①. Design an FSA for bt

step 1 : Understand the pattern

pattern : bt

| part | meaning |
|------|---------|
|------|---------|

b

the first b is compulsory.

+

The preceding b may repeat one (or) more time.

step 2 : check accepted and rejected strings

Accepted strings

b

bb

bbb

...

Why accepted

contains 1b

contains 2b's

contains 3b's

...

Rejected strings

E (Empty)

a

ba

abb

Why rejected

atleast one b required.

the symbol is not b.

contains a, not allowed.

starts with a.

step 3 : Identify the mistake.

Mistake

bt

abt

bbt

meaning.

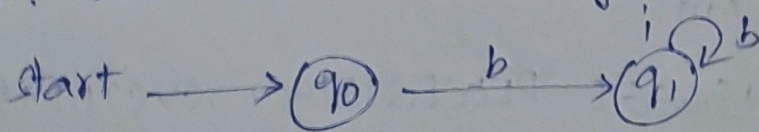
allows zero occurrence (empty).

allows starts with a, not allowed.

extra b before first b.



Step 4: Draw the state diagram.



$q_1$  is the final state.

Step 5: Transition table.

| Current state | Input | Next state |
|---------------|-------|------------|
| $q_0$         | b     | $q_1$      |
| $q_1$         | b     | $q_1$      |

Step 6: Sample inputs

Accepted: b, bb, bbb, ...

Rejected:  $\epsilon$ , a, ba, abb, ...

Conclusion: The FSA accepts all strings having one or more occurrences of b.

② Design an FSA for  $A?B$

Step 1: Understand the pattern

Pattern:  $a?b$

Part meaning

a

The a is optional (zero or one time)

?

It may occur zero or one time

b

The b is compulsory.

Step 2: check accepted and rejected strings

Accepted strings

b

ab

why accepted.

a is absent.

a is present once.



Rejected strings

a

bb

aa

abb

Why rejected.

b is missing.

extra b

b is missing.

extra b after ab.

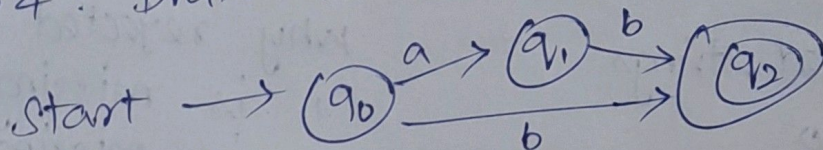
Step 3: Identify the mistake.

a? more than one a not allowed.

ba wrong order.

b? b is optional, not allowed.

Step 4: Draw the state diagram.



Step 5: Transition table

| current state  | Input | Next state     |
|----------------|-------|----------------|
| q <sub>0</sub> | a     | q <sub>1</sub> |
| q <sub>0</sub> | b     | q <sub>2</sub> |
| q <sub>1</sub> | b     | q <sub>2</sub> |

Step 6: Sample inputs.

Accepted : b, ab

Rejected : a, bb, aa, abb

Conclusion : The FSA accepted either b or ab.



③ Design an FSA for 01+ :

Step 1: Understand the pattern.

Pattern

Meaning

0

the first symbol must be 0.

1+

one or more 1's after 0.

Step 2: Check accepted and rejected strings.

Accepted strings

Why accepted

01

Contain one 1

011

Contain two 1's

0111

Contain three 1's

...

Rejected strings

Why rejected

0

1 is missing.

1

0 is missing.

10

wrong order.

010

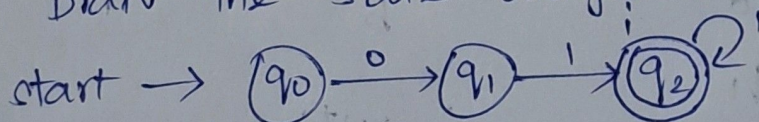
ends with 0, not allowed.

Step 3: Identify the mistake.

10

wrong order.

Step 4: Draw the state diagram.



Step 5: Transition table

| Current state  | Input | Next state     | Explanation.        |
|----------------|-------|----------------|---------------------|
| q <sub>0</sub> | 0     | q <sub>1</sub> | read first 0.       |
| q <sub>1</sub> | 1     | q <sub>2</sub> | read first 1.       |
| q <sub>2</sub> | 1     | q <sub>2</sub> | extra 1's are read. |



step 6 : sample inputs :

Accepted : 01, 011, 0111, 01111.

Rejected : 0, 10, 010.

Conclusion : The FSA accepts strings starting with 0 followed by one or more 1's.

④. Design an FSA for cat / Dog :

step 1 : Understand the pattern.

pattern : cat / dog.

Part

cat

/

dog.

meaning.

accept the word cat.  
or operator.

accept the word dog.

step 2 : check accepted and rejected strings.

Accepted strings

cat

dog

why accepted

matches cat.

matches dog.

Rejected strings

ca

do

cats

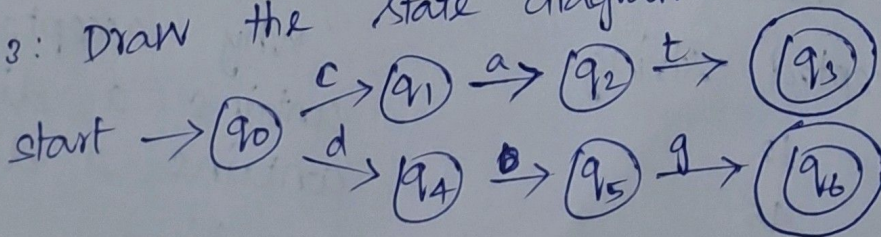
why rejected.

incomplete

incomplete.

extra character.

step 3 : Draw the state diagram.



step 4 : Transition table



| current state | Input | Next state |
|---------------|-------|------------|
| $q_0$         | c     | $q_1$      |
| $q_1$         | a     | $q_2$      |
| $q_2$         | t     | $q_3$      |

| current state | Input | Next state |
|---------------|-------|------------|
| $q_0$         | d     | $q_4$      |
| $q_4$         | o     | $q_5$      |
| $q_5$         | g     | $q_6$      |

step 5 : Sample inputs

Accepted : cat, dog.

Rejected : ca, do, cats, dogs.

Conclusion : The FSA accepts only the words cat or dog.

⑤ Design an FSA for  $[0-9][0-9]$  :

step 1 : Understand the pattern.

Pattern :  $[0-9][0-9]$

Part

$[0-9]$

$[0-9]$

meaning

Any single digit from 0 to 9.

Another digit must follow.

Exactly 2 digits are allowed.

step 2 : check accepted and rejected strings.

Accepted strings

12

45

...

Why accepted

contains 2 digits.

contains 2 digits.

....

Rejected strings

1

123

...

Why rejected.

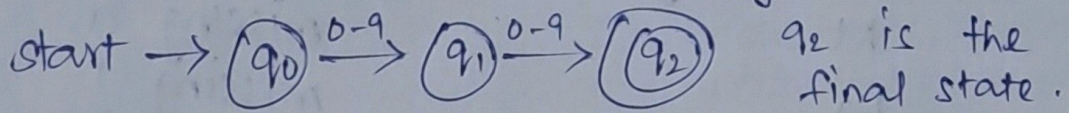
only one digit

more than 2 digits

....



step 3: Draw the state diagram.



step 4: Transition table.

| current state | Input | Next state | Explanation.          |
|---------------|-------|------------|-----------------------|
| $q_0$         | 0-9   | $q_1$      | first digit is read.  |
| $q_1$         | 0-9   | $q_2$      | second digit is read. |

step 5: sample inputs

Accepted: 10, 25, 99, 00

Rejected: 1, 123, a1, 9a.

Conclusion: The FSA accepts exactly 2 digits.

⑥. Design an FSA for hat!

step 1: Understand the pattern.

Pattern: hat!

Part

h

a

!

meaning

first character must be h.

one or more a's.

exclamation mark at the end.

step 2: check accepted & rejected strings.

Accepted strings.

ha!

haa!

haaa!

....

Why accepts.

one a.

two a's

three a's

....



Rejected strings

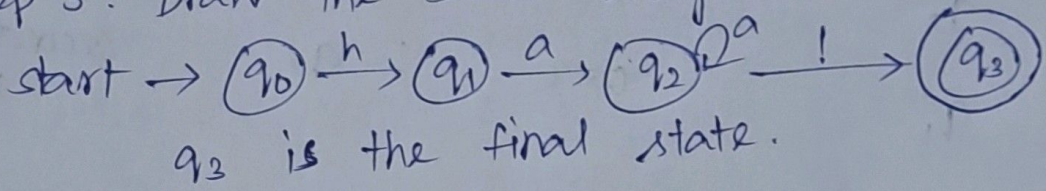
h!  
ha  
haaa  
a!

Why rejected.

missing a.  
missing !  
missing !

does not start with h.

Step 3: Draw the state diagram.



Step 4: Transition table.

| Current state  | Input | Next state     | Explanation.        |
|----------------|-------|----------------|---------------------|
| q <sub>0</sub> | h     | q <sub>1</sub> | read first h.       |
| q <sub>1</sub> | a     | q <sub>2</sub> | read first a.       |
| q <sub>2</sub> | +     | q <sub>2</sub> | extra a's are read. |
| q <sub>2</sub> | !     | q <sub>3</sub> | read ! and accept.  |

Step 5: Sample inputs.

Accepted : ha!, haa!, haaa!

Rejected : h!, ha, haaa, a!

Conclusion : The FSA accepts strings of the form ha n! Where  $n \geq 1$ .

(7) Design an FSA for (01)+.

Step 1: Understand the pattern.

pattern

01  
+

meaning.

The sequence 0 followed by 1.  
one or more repetitions.

Step 2: check accepted and rejected strings.

Accepted strings

01  
0101  
.....

Why accepted

one repetition.  
two repetition.  
.....



Rejected string

0  
1  
011

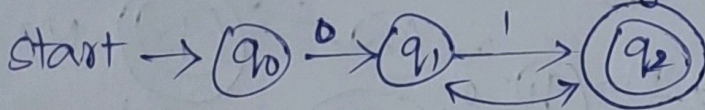
Why rejected.

incomplete sequence.

incomplete sequence..

wrong order.

step 3: Draw the state diagram.



q<sub>2</sub> is the final state.

step 4: Transition table.

| current state  | Input | Next state     | Explanation.           |
|----------------|-------|----------------|------------------------|
| q <sub>0</sub> | 0     | q <sub>1</sub> | first 0 is read.       |
| q <sub>1</sub> | 1     | q <sub>2</sub> | read first 1.          |
| q <sub>2</sub> | 0     | q <sub>1</sub> | next repetition starts |

step 5: Sample inputs:

Accepted: 01, 0101, 010101, ...

Rejected: 0, 1, 011, 011.

Conclusion: The FSA accepts one or more repetitions of 01.

⑧ Design an FSA for Mr/Mrs/Ms.

step 1: Understand the pattern.

Pattern: Mr/Mrs/Ms.

Part

meaning.

Mr

accepts the title Mr.

Mrs

accepts the title Mrs.

Ms

accepts the title Ms.

step 2: check accepted & rejected strings.

Accepted strings

Why accepted.

Mr

accepts the title Mr.

Mrs

accepts the title Mrs.

Ms

matches Ms.



### Rejected strings

M

Mayy

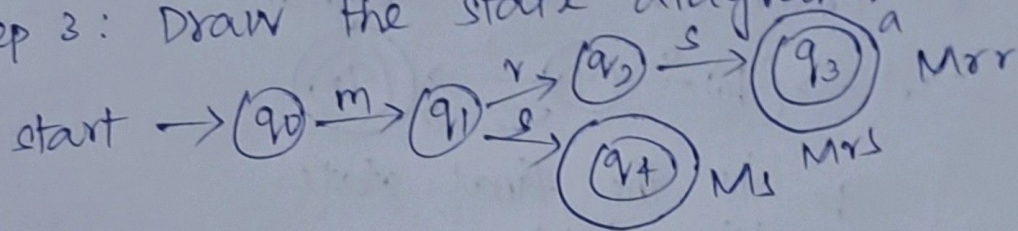
— — —

Why rejected  
incomplete.  
incomplete.

step 3: Draw the state diagram.

```

graph LR
    start(( )) --> q0((q0))
    q0 -- s --> q1(((q1)))
  
```



step 4: Transition table.

| current state | Input | Next state |
|---------------|-------|------------|
| $q_0$         | M     | $q_1$      |
| $q_1$         | r     | $q_2$      |
| $q_2$         | s     | $q_3$      |
| $q_3$         | S     | $q_4$      |

step 5: sample inputs

Accepted : Mr, Mrs, Ms.

Accepted : Mr, Mrs, Miss, Ms.  
Rejected : M, Mr, Mrs, Msr.

Conclusion: The FSA accepts the title Mr, Mrs, Ms.